Name : A Syed Thoufiq

Reg No : 192424211

Course/Code : DSA0604 - DATA HANDLING AND VISUALIZATION

Date : 15/07/26

QUESTION 1: SMART CITY AIR QUALITY MONITORING

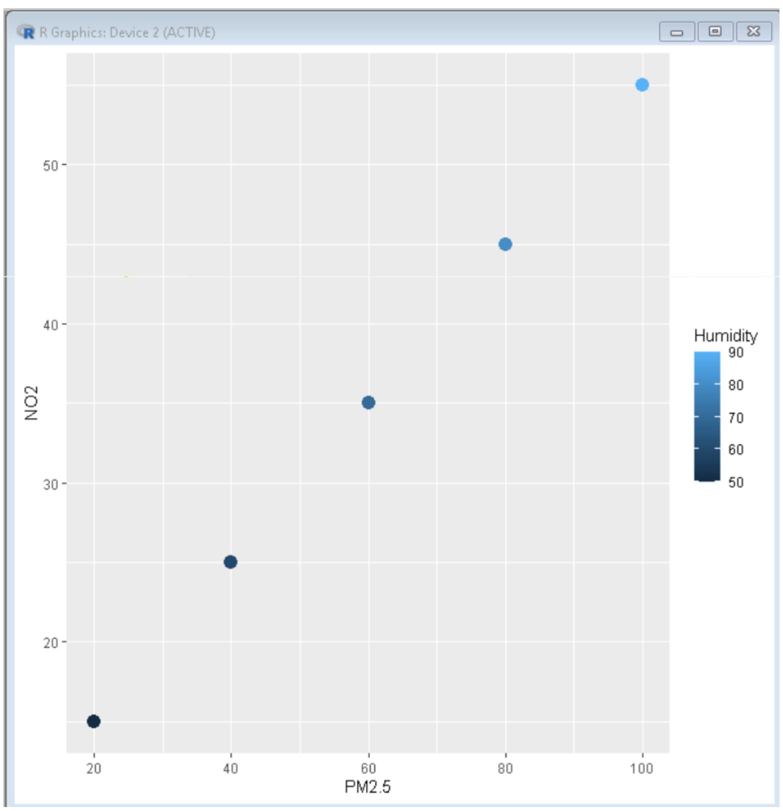
CODE:

```
library(ggplot2)

air <- data.frame(
  PM2.5=c(20,40,60,80,100),
  NO2=c(15,25,35,45,55),
  Humidity=c(50,60,70,80,90)
)

ggplot(air, aes(PM2.5, NO2, color=Humidity)) +
  geom_point(size=4)
```

OUTPUT:



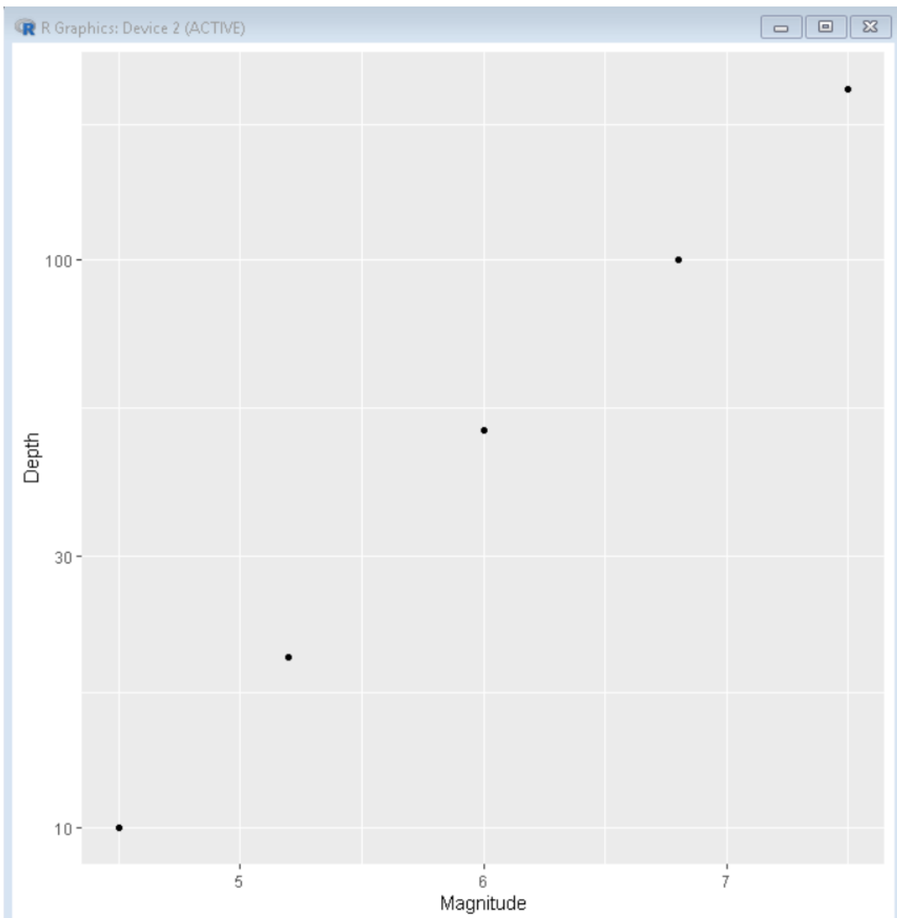
QUESTION 2: EARTHQUAKE RISK ANALYSIS

CODE:

```
earthquake <- data.frame(
  Magnitude=c(4.5,5.2,6.0,6.8,7.5),
  Depth=c(10,20,50,100,200)
)

ggplot(earthquake, aes(Magnitude, Depth)) +
  geom_point() +
  scale_y_log10()
```

OUTPUT:



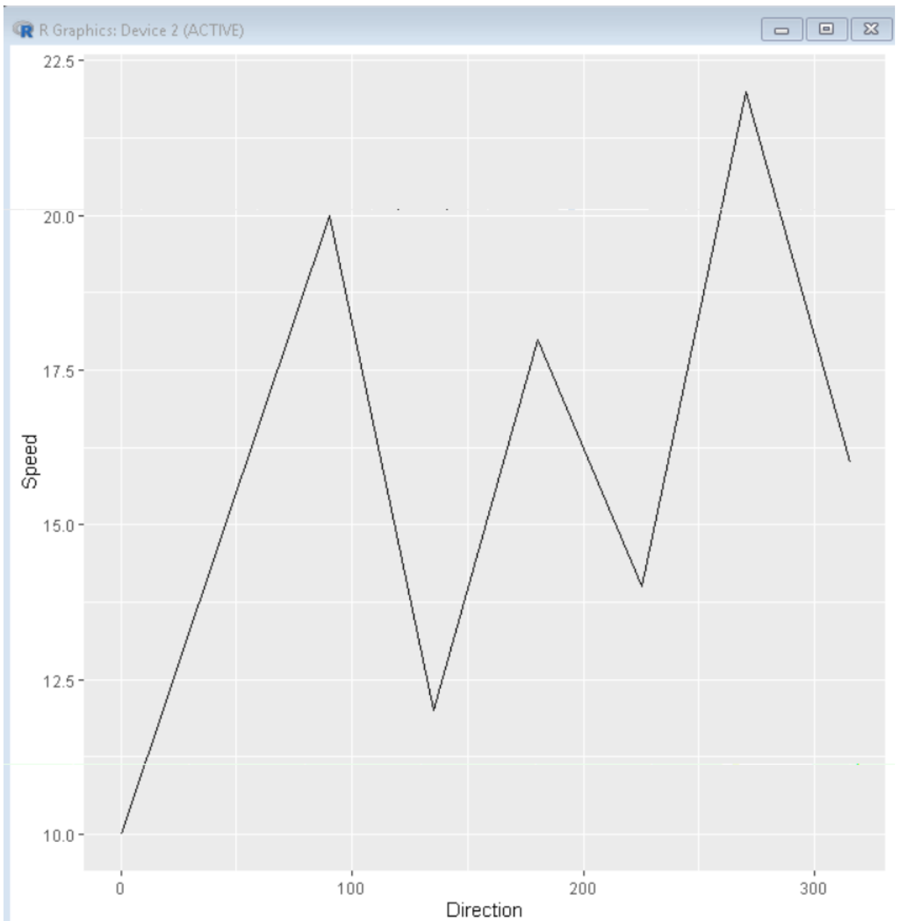
QUESTION 3: WIND DIRECTION ANALYSIS

CODE:

```
wind <- data.frame(
  Direction=c(0,45,90,135,180,225,270,315),
  Speed=c(10,15,20,12,18,14,22,16)
)

ggplot(wind, aes(Direction, Speed)) +
  geom_line()
```

OUTPUT:



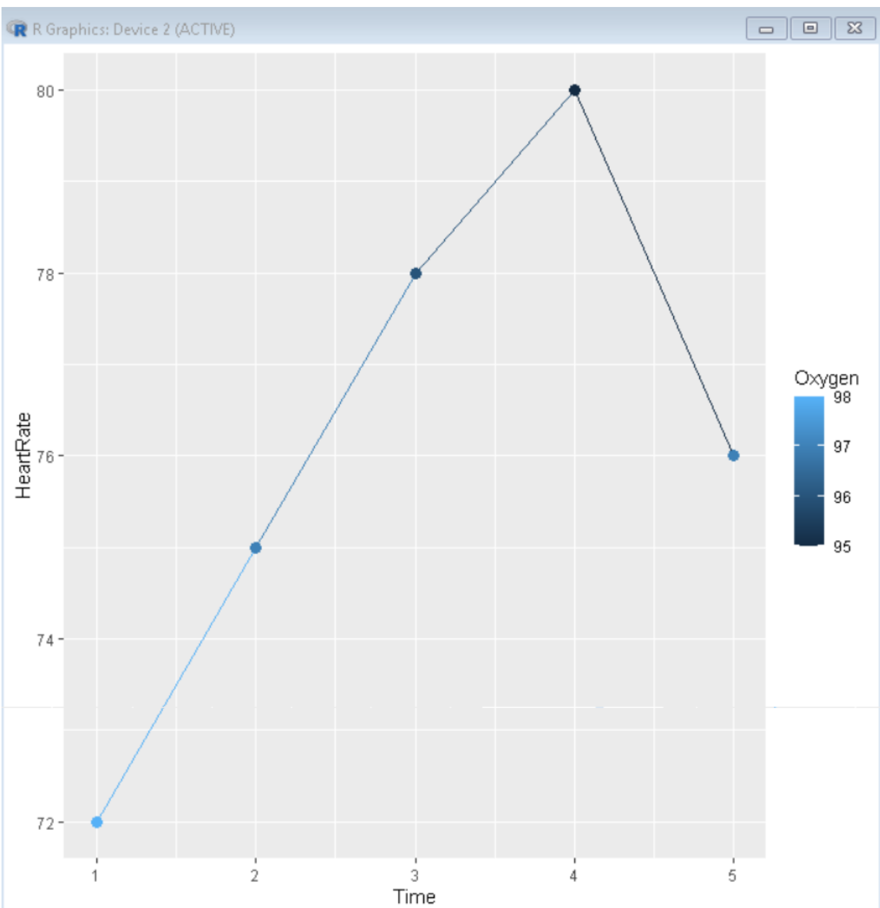
QUESTION 4: HOSPITAL PATIENT MONITORING

CODE:

```
patient <- data.frame(
  Time=1:5,
  HeartRate=c(72,75,78,80,76),
  Oxygen=c(98,97,96,95,97)
)

ggplot(patient, aes(Time, HeartRate, color=Oxygen)) +
  geom_line() +
  geom_point(size=3)
```

OUTPUT:



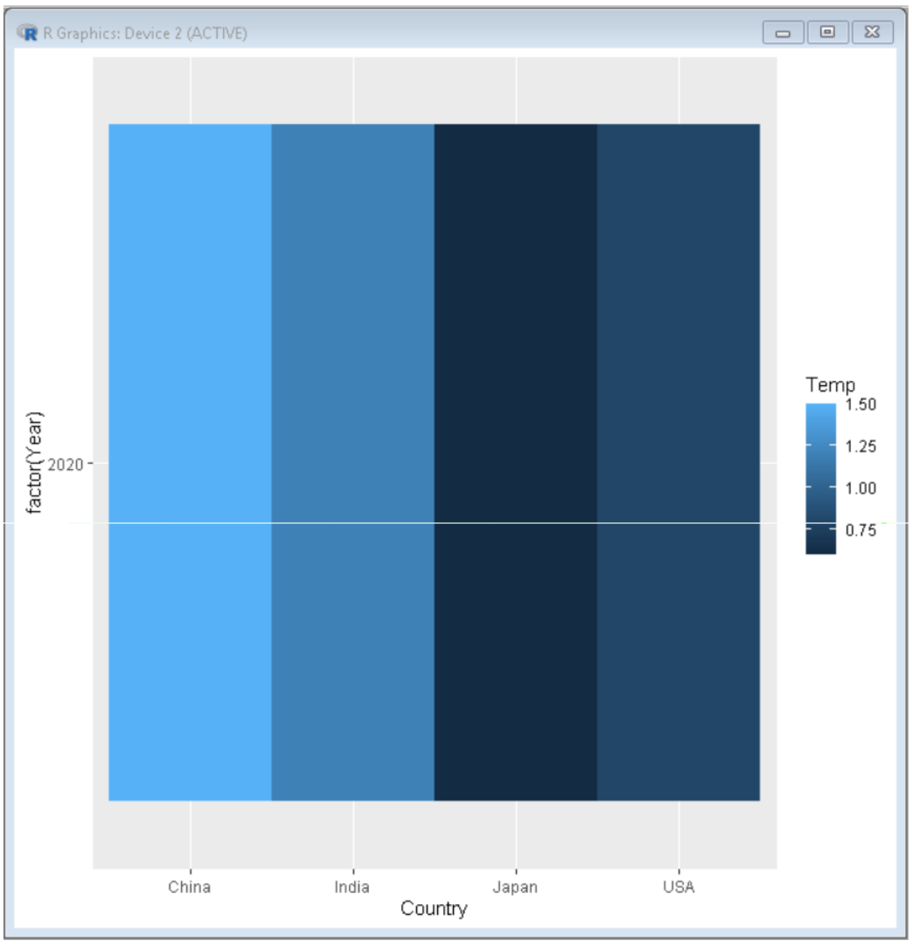
QUESTION 5: GLOBAL CLIMATE CHANGE STUDY

CODE:

```
climate <- data.frame(
  Country=c("India","USA","China","Japan"),
  Year=c(2020,2020,2020,2020),
  Temp=c(1.2,0.8,1.5,0.6)
)

ggplot(climate, aes(Country, factor(Year), fill=Temp)) +
  geom_tile()
```

OUTPUT:



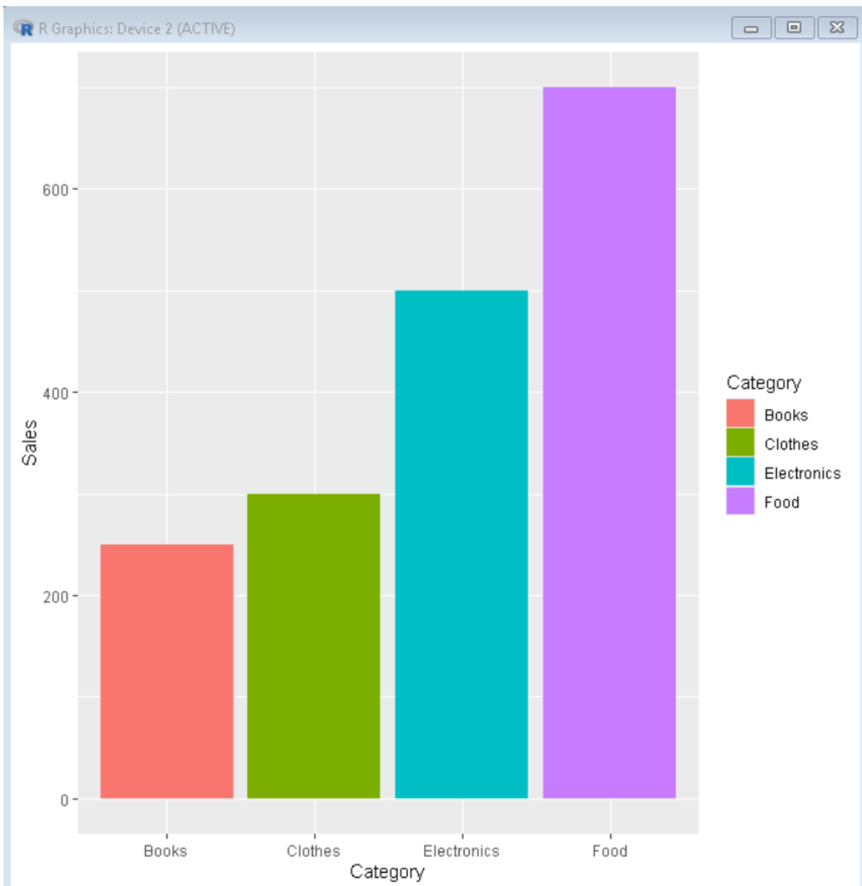
QUESTION 6: E-COMMERCE SALES ANALYSIS

CODE:

```
sales <- data.frame(
  Category=c("Electronics","Clothes","Food","Books"),
  Sales=c(500,300,700,250)
)

ggplot(sales, aes(Category, Sales, fill=Category)) +
  geom_bar(stat="identity")
```

OUTPUT:



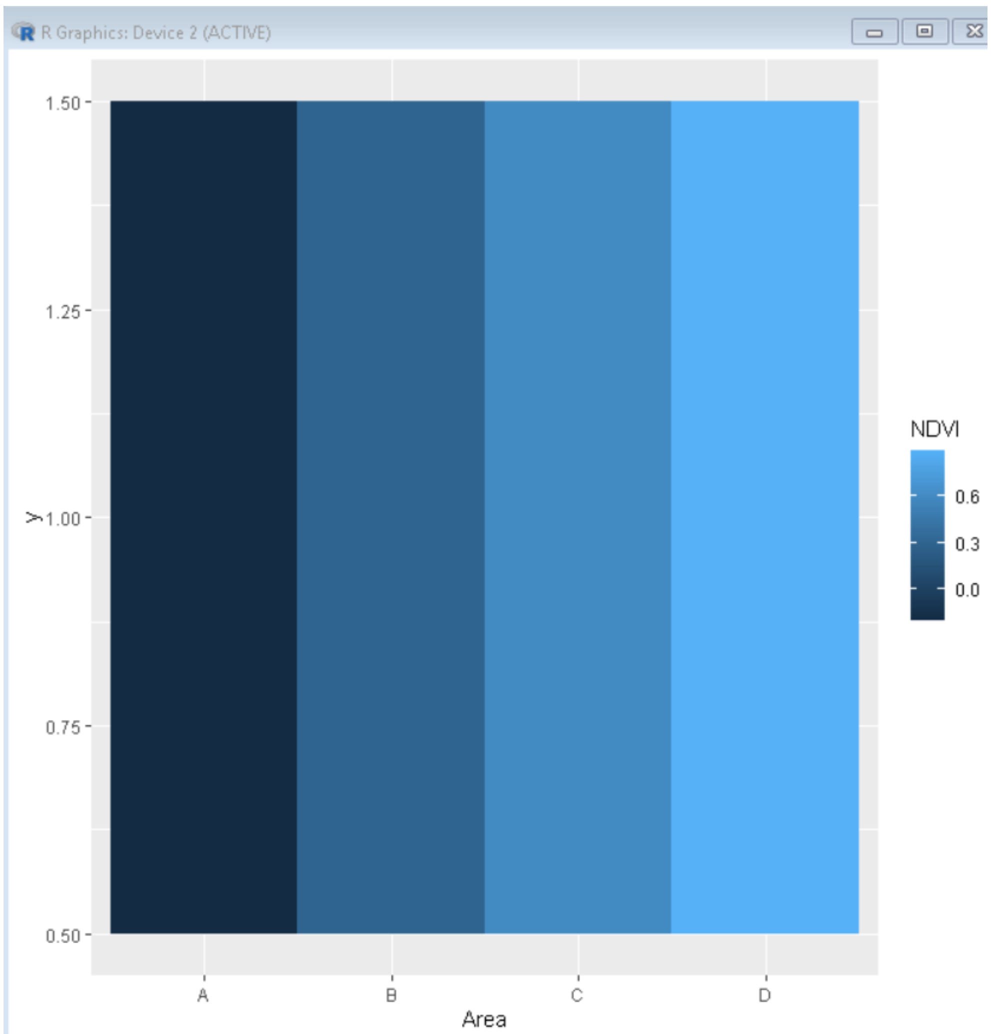
QUESTION 7: SATELLITE IMAGE VEGETATION ANALYSIS

CODE:

```
ndvi <- data.frame(
  Area=c("A","B","C","D"),
  NDVI=c(-0.2,0.3,0.6,0.9)
)

ggplot(ndvi, aes(Area, 1, fill=NDVI)) +
  geom_tile()
```

OUTPUT:



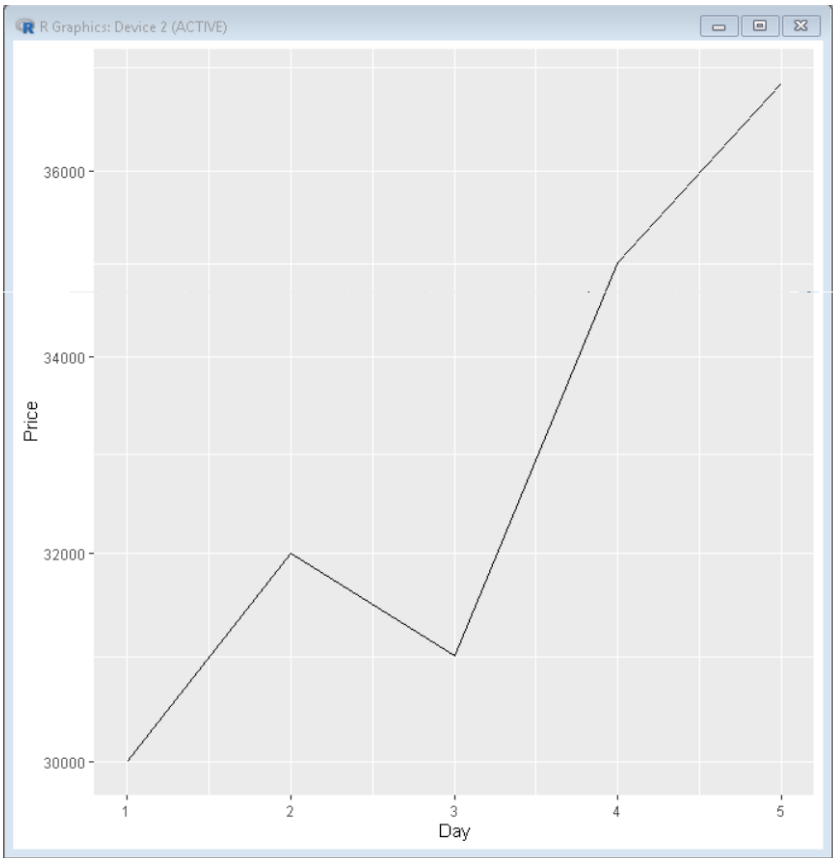
QUESTION 8: CRYPTO CURRENCY MARKET VOLATILITY

CODE:

```
crypto <- data.frame(
  Day=1:5,
  Price=c(30000,32000,31000,35000,37000)
)

ggplot(crypto, aes(Day, Price)) +
  geom_line() +
  scale_y_log10()
```

OUTPUT:



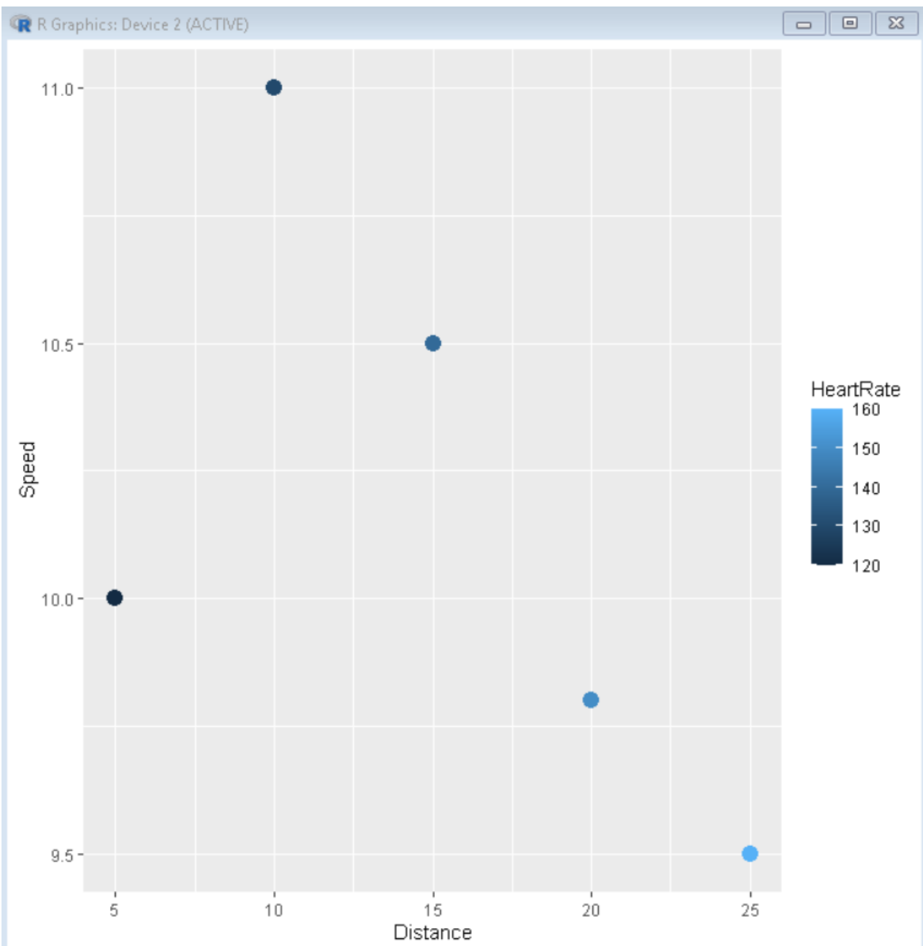
QUESTION 9: MARATHON RUNNER PERFORMANCE

CODE:

```
runner <- data.frame(
  Distance=c(5,10,15,20,25),
  Speed=c(10,11,10.5,9.8,9.5),
  HeartRate=c(120,130,140,150,160)
)

ggplot(runner, aes(Distance, Speed, color=HeartRate)) +
  geom_point(size=4)
```

OUTPUT:



QUESTION 10: COVID-19 VACCINATION CAMPAIGN DASHBOARD

Code:

```
covid <- data.frame(
  District=c("A","B","C","D"),
  Vaccination=c(65,80,90,75)
)

ggplot(covid, aes(District, Vaccination, fill=Vaccination)) +
  geom_bar(stat="identity")
```

OUTPUT: